

# Week 1 – Introductory Concepts

## 1.1 Definition of a Computer Network

A **computer network** is a system of interconnected devices, such as computers, printers, smartphones, servers, and sensors, that are linked together for the purpose of sharing resources and communicating with one another. Unlike a standalone computer, which can only process and store information for a single user, a network enables collaboration, information exchange, and access to centralized services.

For example, in a university network, students can access shared printers, download lecture notes from a central server, and communicate with lecturers through email — all because their devices are connected within the same network.

The key idea is that a network turns isolated devices into participants of a larger, cooperative system.

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## 1.2 Components of Data Communication

For communication to occur between two devices, several essential components must be present. These are:

1. **Message** – This is the actual information to be transmitted. It can take many forms: plain text in an email, an image, a video stream, or even sensor data from IoT devices.
2. **Sender** – The device that generates and sends the message. This could be a laptop sending an email, a smartphone uploading a video, or even a server transmitting data to clients.
3. **Receiver** – The target device that accepts and interprets the message. For example, when you send a WhatsApp message, your phone is the sender and your friend's phone is the receiver.
4. **Transmission Medium** – The physical path that carries the data from sender to receiver. Examples include copper cables (Ethernet), fiber optic cables (light transmission), and wireless signals (radio waves, microwaves, infrared).
5. **Protocol** – A protocol is a set of rules that define how communication takes place. Without protocols, devices would not “speak the same language.” Examples include TCP/IP for Internet communication, HTTP for web browsing, and SMTP for email.

🔑 **Key point:** If any of these five components is missing, communication will either fail or be incomplete.

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## 1.3 Importance of Networks

Computer networks are essential to modern life and business. Their importance can be understood through the following aspects:

- **Resource Sharing:** Instead of each employee in an office having a personal printer, a single printer can be shared across the entire network. Similarly, files stored on a central server can be accessed by multiple users.
  - **Collaboration:** Networks make remote collaboration possible. For example, Google Docs allows multiple users to edit a single document in real time from different parts of the world.
  - **Cost Efficiency:** Networks reduce duplication of resources. Centralized servers or cloud storage are cheaper and more efficient than equipping every user with large amounts of storage.
  - **Reliability & Redundancy:** With backup servers and redundant paths, networks ensure availability even if part of the system fails.
  - **Challenges:** At the same time, networks introduce risks such as security threats (hackers, malware), privacy concerns (data leaks), and vulnerabilities to failure (a single point of failure in poorly designed networks can affect many users).
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## 1.4 Data Representation in Networks

Since computers work with binary (0s and 1s), all forms of data must be represented in a format that can be stored and transmitted digitally.

- **Text:** Represented using standard codes. The most common are ASCII (American Standard Code for Information Interchange) and Unicode (which supports a wide range of characters, including non-English scripts). For example, the letter “A” is 65 in ASCII and 01000001 in binary.
  - **Numbers:** Directly represented in binary. For human readability, they are often converted into decimal (base 10) or hexadecimal (base 16).
  - **Images:** Represented as a grid of pixels, where each pixel has a color value. Compression techniques like JPEG or PNG are used to reduce file size without significantly reducing quality.
  - **Audio:** Sound is originally analog (continuous waves). To transmit it digitally, the sound wave is sampled at intervals and converted into binary (e.g., MP3, WAV).
  - **Video:** A video is essentially a series of images (frames) played rapidly, often 24–60 frames per second. Compression standards like MPEG or MP4 make video streaming practical over networks.
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## 1.5 Line Configurations (Ways of Connecting Devices)

Devices in a network can be physically connected in different ways:

- **Point-to-Point:** A direct dedicated link between two devices. Example: a PC connected directly to a printer with a cable. This configuration is simple but not scalable.
  - **Multipoint:** A single link is shared among multiple devices. For instance, in an older bus topology LAN, all computers connect to the same coaxial cable. While cost-efficient, this can lead to performance issues if many devices transmit simultaneously.
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## 1.6 Transmission Modes

The way data flows between sender and receiver can vary:

- **Simplex:** Communication is one-way only. For example, a traditional radio broadcast — listeners receive signals but cannot send anything back.
  - **Half-Duplex:** Communication can flow both ways, but only one direction at a time. Walkie-talkies are a good example; when one person speaks, the other must wait before responding.
  - **Full-Duplex:** Communication can occur in both directions simultaneously. A telephone conversation allows both parties to speak and listen at the same time.
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## 1.7 Performance Metrics

The effectiveness of a network can be measured using several metrics:

- **Bandwidth:** The theoretical maximum capacity of a communication link, usually measured in bits per second (bps). For example, a fiber optic link might support 1 Gbps.
- **Throughput:** The actual rate of successful data delivery. Often lower than bandwidth due to congestion, errors, or overhead.
- **Latency:** The time delay for data to travel from sender to receiver, measured in milliseconds. Critical for real-time applications like gaming or video conferencing.
- **Jitter:** The variation in latency between packets. High jitter disrupts streaming and voice calls.

💡 **Example:** A 4K Netflix stream requires high throughput (for video quality), low latency (for smooth streaming), and low jitter (to prevent interruptions).

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## 1.8 Standards and Protocols

Without universal standards, networks would be chaotic because devices from different manufacturers would not interoperate. Standards ensure compatibility.

- **IEEE (Institute of Electrical and Electronics Engineers):** Defines LAN/WLAN standards like Ethernet (802.3) and Wi-Fi (802.11).
- **IETF (Internet Engineering Task Force):** Defines Internet standards like TCP/IP and HTTP.
- **ISO (International Organization for Standardization):** Developed the OSI model.
- **ITU (International Telecommunication Union):** Sets global telecom and radio communication standards.

Protocols are specific sets of rules. For example, **HTTP** defines how web browsers and servers exchange web pages, while **SMTP** defines how emails are sent.

## 2.1 Types of Networks by Scope

Networks can be classified according to their geographical coverage and scale.

- **Personal Area Network (PAN):**  
A very small network, typically within 10 meters, that connects personal devices such as smartphones, tablets, and Bluetooth headsets. Example: Wireless connection between a smartwatch and a mobile phone.
  - **Local Area Network (LAN):**  
Covers a small area like a building or campus. It is privately owned, fast, and cost-effective. Example: A company's office LAN connecting employees' computers, printers, and servers.
  - **Campus Area Network (CAN):**  
Connects multiple LANs within a limited area such as a university or hospital campus. Larger than a LAN but smaller than a MAN. Example: A university linking dormitories, lecture halls, and administrative buildings.
  - **Metropolitan Area Network (MAN):**  
Spans a city or metropolitan area. Often provided by telecom companies. Example: A city-wide fiber optic network used by government offices.
  - **Wide Area Network (WAN):**  
Covers large geographical regions, often across countries and continents. The Internet is the largest WAN. Uses leased lines, satellites, or undersea cables.
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## 2.2 Internet, Intranet, Extranet

- **Internet:** A public, global network of networks, accessible to anyone with a device and ISP connection.
  - **Intranet:** A private, internal network within an organization, accessible only by employees. Provides internal services like company announcements and file sharing.
  - **Extranet:** An intranet extended to trusted external users such as suppliers or business partners. Example: A logistics company giving suppliers access to its inventory system.
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## 2.3 Communication Media

**Guided (Wired Media):**

- **Twisted Pair Cables:** Two copper wires twisted together to reduce electromagnetic interference. Categories (Cat5e, Cat6, Cat6A) define bandwidth and speed. Most common LAN medium.
- **Coaxial Cable:** Consists of a central copper conductor with shielding. Used for cable TV and early Ethernet.
- **Fiber Optic Cable:** Uses light to transmit data, immune to electromagnetic interference, supports extremely high bandwidth over long distances.

#### Unguided (Wireless Media):

- **Radio Waves:** Used in Wi-Fi, Bluetooth, and mobile networks. Provides flexibility but is subject to interference.
  - **Microwave:** Line-of-sight communication, often used for long-distance links.
  - **Satellite:** Covers very large distances, useful for remote areas but suffers from high latency.
  - **Infrared:** Very short-range, often used in remote controls and device-to-device transfers.
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## 2.4 Enterprise Network Architecture

Large organizations design their networks in **layers** for efficiency and scalability:

- **Access Layer:** The edge of the network, where end devices such as computers and phones connect (via switches and wireless access points).
  - **Distribution Layer:** Aggregates access layer connections, applies policies such as security and quality of service (QoS).
  - **Core Layer:** The backbone of the network, providing high-speed, reliable switching and routing between distribution layers.
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## 2.5 Cabling Infrastructure in Enterprises

- **MDF (Main Distribution Frame):** The central point that connects the internal network to the outside world (e.g., ISP).
- **IDF (Intermediate Distribution Frame):** Secondary distribution points located in different floors or buildings.
- **Horizontal Cabling:** Runs from the IDF to end-user devices.
- **Backbone Cabling:** Connects MDF and IDFs to ensure connectivity across the enterprise.

### 3.1 Definition of a LAN

A **Local Area Network (LAN)** is a network that connects devices within a relatively small geographical area such as a building, office, or campus. LANs provide **high data transfer rates** and are usually privately owned and managed.

Example: A university computer lab LAN allows students to share files, access the Internet, and print documents from any workstation.

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### 3.2 LAN Topologies

#### Physical vs Logical Topology

- **Physical Topology:** The actual physical layout of cables and devices.
- **Logical Topology:** The way data flows within the network regardless of physical layout.

#### Common Topologies:

- **Bus:** All devices share a single backbone cable. Simple but prone to collisions and failures if the backbone is damaged.
  - **Star:** Devices connect to a central hub or switch. The most widely used topology today because it is easy to manage. If one cable fails, only one device is affected.
  - **Ring:** Devices connected in a closed loop. Data travels in one direction around the ring. Faults in the ring can affect the whole network unless dual rings are used.
  - **Mesh:** Every device connects directly to every other device. Offers maximum redundancy but is expensive and complex.
  - **Hybrid:** Combination of two or more topologies. Example: A star-bus topology in a large office.
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### 3.3 Transmission in LANs

- **Baseband Transmission:** Uses the entire medium to send one signal at a time (Ethernet). Digital in nature.
  - **Broadband Transmission:** Allows multiple signals to travel simultaneously over the same medium (like cable TV networks).
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### 3.4 Access Methods

When many devices share the same communication medium, an **access method** determines who can transmit at a given time.

- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):**  
Used in early Ethernet. Devices “listen” to the medium before transmitting. If two transmit simultaneously, a collision occurs. Both devices stop, wait a random time, and retry.
  - **Token Passing:**  
A token (special frame) circulates on the network. Only the device holding the token may transmit. Used in Token Ring networks. Eliminates collisions but adds overhead.
  - **Polling:**  
A central controller gives permission to each device in turn to transmit. Ensures fairness but introduces delay.
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### 3.5 Ethernet – The Standard LAN Technology

- Standardized by IEEE 802.3.
  - Uses MAC addresses to identify devices.
  - Frame structure includes preamble, source and destination MAC addresses, type/length, payload, and error-checking field (CRC).
  - Speeds range from 10 Mbps (legacy) up to 400 Gbps (modern data centers).
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### 3.6 Wireless LANs (Wi-Fi)

- Standardized by IEEE 802.11 (with variants a/b/g/n/ac/ax).
  - Uses **CSMA/CA (Collision Avoidance)** instead of CSMA/CD. Devices attempt to avoid collisions by waiting before transmitting.
  - **Access Points (APs)** act as central hubs in wireless LANs.
  - **BSS (Basic Service Set):** One AP with associated devices.
  - **ESS (Extended Service Set):** Multiple APs interconnected, allowing roaming across larger areas.
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### 3.7 Comparison: Wired LAN vs Wireless LAN

- Wired LANs offer higher speed, security, and reliability.

- Wireless LANs provide mobility and ease of installation but are prone to interference and security risks.
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## Week 4 – Network Devices I (NICs, Hubs, Bridges, Switches)

### 4.1 Network Interface Card (NIC)

- The NIC is the hardware that enables a device to connect to a network.
  - Each NIC has a unique **MAC address**.
  - Responsible for framing, error detection, and media access.
  - Can be wired (Ethernet NIC) or wireless (Wi-Fi NIC).
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### 4.2 Repeaters and Hubs

- **Repeater:** Amplifies signals to extend transmission distance. Does not interpret data, simply regenerates it. Operates at Layer 1.
  - **Hub:** A multiport repeater. Broadcasts data to all ports, regardless of destination. Creates one large collision domain. Obsolete in modern LANs.
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### 4.3 Bridges

- Operate at Layer 2.
  - Inspect MAC addresses to decide whether to forward or filter traffic.
  - Reduce collisions by segmenting networks.
  - All segments remain part of one broadcast domain.
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### 4.4 Switches

- Multiport bridges that forward frames intelligently.
- Learn MAC addresses and build a switching table.
- Provide dedicated bandwidth to each port.
- Each port is its own collision domain, improving performance.

#### Switching Methods:

- Store-and-Forward (checks entire frame, error detection).

- Cut-Through (forwards immediately after reading MAC address).
  - Fragment-Free (waits for first 64 bytes to avoid collision fragments).
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## 4.5 Broadcast and Collision Domains

- **Collision Domain:** The area where collisions may occur. One per switch port.
  - **Broadcast Domain:** The area where broadcast traffic is delivered. Entire switch unless VLANs or routers are used.
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## 4.6 Spanning Tree Protocol (STP)

- Prevents loops in networks with redundant paths.
  - Ensures a loop-free logical topology by disabling some links temporarily.
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## 4.7 Power over Ethernet (PoE)

- Supplies both power and data through Ethernet cables.
- Useful for IP phones, wireless access points, and security cameras.

## Week 5 – Network Devices II (Routers, Routing, NAT, ACLs)

### 5.1 Routers

A **router** is a networking device that forwards data packets between different networks based on their IP addresses.

- Operates at **Layer 3 (Network Layer)** of the OSI model.
- Maintains a **routing table** that lists known networks and the next hop to reach them.
- Unlike switches, routers separate both **collision domains** and **broadcast domains**, improving efficiency.

**Example:** A home router connects your private LAN (192.168.x.x) to the public Internet.

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### 5.2 Static vs Dynamic Routing

- **Static Routing:**

- Configured manually by an administrator.
    - Predictable and secure but does not adapt to failures.
    - Suitable for small networks.
  - **Dynamic Routing:**
    - Uses protocols to discover routes automatically.
    - Adapts to network changes.
    - Examples:
      - *RIP* (Routing Information Protocol, distance vector, uses hop count).
      - *OSPF* (Open Shortest Path First, link-state, uses full network map).
      - *EIGRP* (Cisco proprietary, hybrid).
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### 5.3 Default Gateway

- A host's **default gateway** is the router interface that forwards packets destined for outside its local network.
  - Without it, a device can only communicate within its LAN.
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### 5.4 NAT (Network Address Translation)

NAT allows multiple private IP addresses to share a smaller pool of public IP addresses.

- Solves IPv4 address exhaustion.
- Provides a level of security by hiding internal addresses.

#### Types of NAT:

- **Static NAT:** One private ↔ one public mapping.
  - **Dynamic NAT:** Multiple private addresses share a pool of public addresses.
  - **PAT (Port Address Translation):** Many private addresses share a single public IP, differentiated by port numbers. (Most common in homes and small businesses.)
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### 5.5 ACLs (Access Control Lists)

ACLs are rule sets on routers/firewalls that filter traffic based on conditions.

- Criteria: source/destination IP, port numbers, protocol type.
- Example: Block traffic from 192.168.1.50 to the Internet, allow all else.
- Two types:

- *Standard ACLs* (filter by source address only).
  - *Extended ACLs* (filter by source, destination, protocol, ports).
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## 5.6 Routers vs Layer 3 Switches

- **Router:** More versatile, supports WAN protocols, better for connecting geographically distant networks.
  - **Layer 3 Switch:** Fast hardware-based routing, best for high-speed LAN environments.
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## Week 6 – Network Models (OSI and TCP/IP)

### 6.1 Purpose of Models

Network models divide communication into layers to simplify design, allow vendor interoperability, and make troubleshooting easier.

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### 6.2 OSI Model (7 Layers)

1. **Physical:** Transmission of raw bits (cables, hubs, voltages).
  2. **Data Link:** Framing, error detection, MAC addressing (switches, NICs).
  3. **Network:** Logical addressing and routing (IP, routers).
  4. **Transport:** Reliable delivery, segmentation, flow control (TCP, UDP).
  5. **Session:** Establishing and maintaining communication sessions.
  6. **Presentation:** Data formatting, compression, encryption (SSL, JPEG).
  7. **Application:** End-user services (HTTP, SMTP, DNS, FTP).
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### 6.3 TCP/IP Model (4 Layers)

1. **Network Access:** Physical + Data Link (Ethernet, Wi-Fi).
  2. **Internet:** IP, ICMP, ARP.
  3. **Transport:** TCP and UDP.
  4. **Application:** End-user protocols (HTTP, SMTP, DNS).
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### 6.4 Encapsulation

When data moves down the layers, it is encapsulated with headers (and sometimes trailers).

- Application data → TCP segment → IP packet → Ethernet frame → Bits.  
At the receiving end, **de-encapsulation** occurs in reverse.
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## 6.5 Protocol Data Units (PDUs)

- **Layer 4 (Transport):** Segment (TCP) / Datagram (UDP).
  - **Layer 3 (Network):** Packet.
  - **Layer 2 (Data Link):** Frame.
  - **Layer 1 (Physical):** Bits.
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## 6.6 Ports and Sockets

- Ports identify specific services on a device.
  - Examples: HTTP (80), HTTPS (443), DNS (53).
  - A **socket** = IP address + port number.
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## Week 7 – Mid-Semester Review

This week consolidates concepts learned in Weeks 1–6.

### 7.1 Devices Recap

- NIC – Layer 2, MAC addresses.
- Hub – Layer 1, single collision domain.
- Switch – Layer 2, one collision domain per port.
- Router – Layer 3, separates broadcast domains.
- Gateway – Translates between protocols.

### 7.2 LAN Recap

- Ethernet – CSMA/CD, switched Ethernet.
- Wi-Fi – CSMA/CA, APs, roaming.
- VLANs – Create logical segmentation.

### 7.3 OSI/TCP Recap

- OSI = 7-layer theoretical model.
- TCP/IP = 4-layer practical model.
- Encapsulation = Wrapping data with headers.

## 7.4 Practice Questions

1. Which device separates collision domains but not broadcast domains?
    - Switch.
  2. Which OSI layer is responsible for encryption and compression?
    - Presentation.
  3. What is the main difference between TCP and UDP?
    - TCP is reliable, connection-oriented; UDP is faster, connectionless.
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## Week 8 – IP Addressing

### 8.1 IPv4 Basics

- 32-bit addresses in dotted decimal (e.g., 192.168.1.1).
- Composed of **network ID** and **host ID**.

### 8.2 Address Classes (Legacy)

- Class A: 0–127 (/8).
  - Class B: 128–191 (/16).
  - Class C: 192–223 (/24).
  - Class D: 224–239 (Multicast).
  - Class E: 240–255 (Experimental).
- Modern networks use **CIDR** instead.

### 8.3 Special IPv4 Addresses

- Private: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- Loopback: 127.0.0.0/8.
- APIPA: 169.254.0.0/16.

### 8.4 IPv6 Basics

- 128-bit hexadecimal addresses.
- Virtually unlimited space.
- Supports autoconfiguration, built-in security, no NAT needed.
- Address types: unicast, multicast, anycast.

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## Week 9 – Subnetting

### 9.1 Why Subnet?

- Efficient use of IP addresses.
- Smaller broadcast domains.
- Better performance and security.

### 9.2 Subnetting Example

Given 192.168.1.100/26:

- Mask = 255.255.255.192.
- Block size = 64.
- Subnets: 192.168.1.0, .64, .128, .192.
- 192.168.1.100 belongs to 192.168.1.64 subnet.
- Usable range: .65 – .126.

### 9.3 Subnetting Formulas

- Subnets =  $2^n$  (n = borrowed bits).
- Hosts =  $2^h - 2$  (h = host bits).

### 9.4 CIDR

- Classless addressing allows flexible subnet sizes.
- Example: 10.0.0.0/22 → covers 10.0.0.0 – 10.0.3.255.

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## Week 10 – VLSM (Variable Length Subnet Masking)

### 10.1 What is VLSM?

- Allows subnetting with different mask lengths in the same network.
- Optimizes address space usage.

### 10.2 VLSM Example

Given 192.168.1.0/24 with needs:

- Subnet A (100 hosts) → /25 → 192.168.1.0/25.

- Subnet B (50 hosts) → /26 → 192.168.1.128/26.
- Subnet C (25 hosts) → /27 → 192.168.1.192/27.

### **10.3 Route Summarization**

- Aggregating contiguous subnets into one advertisement.
- Example: 192.168.0.0/24 + 192.168.1.0/24 → summarized as 192.168.0.0/23.

### **10.4 VLSM in IPv6**

- Not required due to enormous address space.
- ISPs typically allocate /48 or /56 per customer.